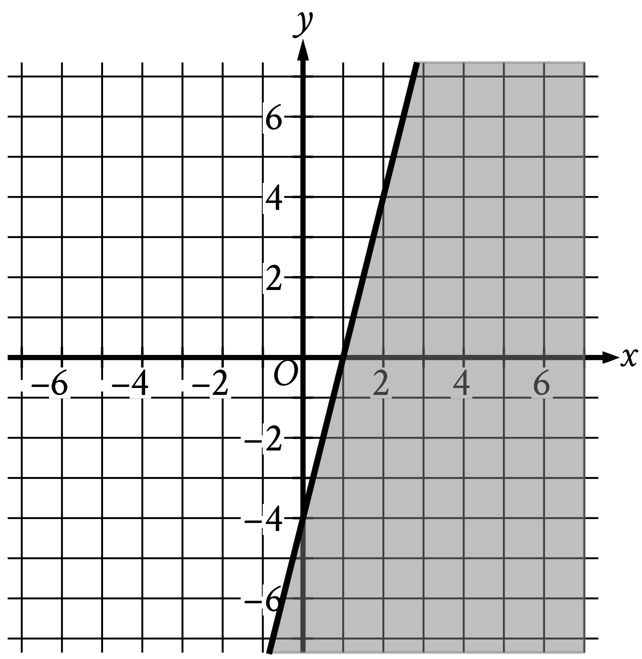
MEDIUM ALGEBRA

Linear inequalities in one or two variables

01

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29



- The boundary of the inequality is a solid line.
    - The line slants sharply up from left to right.
    - The line passes through the following points:
        - (0 comma negative 4)
        - (1 comma 0)
        - (2 comma 4)
- The area below and to the right of the boundary is shaded.

The shaded region shown represents the solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

A. -5, -6

B. -2, 5

C. 1, 4

D. 6, - 2

Q2

ALGEBRA

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall plant species?

A. $106 < h < 158$

B. $80 < h < 100$

C. $42 < h < 87$

D. $17 < h < 85$

Q3

ALGEBRA

The length of a rectangle is 50 inches and the width is x inches. The perimeter is at most 210 inches. Which inequality represents this situation?

A. $2x + 100 \leq 210$

B. $2x + 100 \geq 210$

C. $2x + 50 \leq 210$

D. $2x + 50 \geq 210$

Q4 ALGEBRA

In North America, the standard width of a parking space is at least 7.5 feet and no more than 9.0 feet. A restaurant owner recently resurfaced the restaurant's parking lot and wants to determine the number of parking spaces, n, in the parking lot that could be placed perpendicular to a curb that is 135 feet long, based on the standard width of a parking space. Which of the following describes all the possible values of n ?

A. $18 \leq n \leq 135$

B. $7.5 \leq n \leq 9$

C. $15 \leq n \leq 135$

D. $15 \leq n \leq 18$

Q5 ALGEBRA

$$y > 7x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	13
5	27
8	48

B.

x	y
3	17
5	31
8	52

C.

x	y
3	21
5	27
8	52

D.

x	y
3	21
5	35
8	56

Q6

ALGEBRA

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than 164 total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where a is the number of azaleas that will be planted, and b is the number of boxwoods that will be planted?

A. $a + b \geq 164$
$3a \geq b$

B. $a + b \geq 164$
$a \leq 3b$

C. $a + b \leq 164$
$3a \geq b$

D. $a + b \leq 164$
$a \leq 3b$

Q7

GRID-IN ALGEBRA

An event planner is planning a party. It costs the event planner a onetime fee of $\$ 35$ to rent the venue and $\$ 10.25$ per attendee. The event planner has a budget of $\$ 200$. What is the greatest number of attendees possible without exceeding the budget?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ALGEBRA

A certain elephant weighs 200 pounds at birth and gains more than 2 but less than 3 pounds per day during its first year. Which of the following inequalities represents all possible weights w, in pounds, for the elephant 365 days after birth?

A. $400 < w < 600$

B. $565 < w < 930$

C. $730 < w < 1,095$

D. $930 < w < 1,295$

Q9

ALGEBRA

The minimum value of x is 12 less than 6 times another number n. Which inequality shows the possible values of x?

A. $x \leq 6n - 12$

B. $x \geq 6n - 12$

C. $x \leq 12 - 6n$

D. $x \geq 12 - 6n$

Q10

ALGEBRA

The average annual energy cost for a certain home is $4,334. The homeowner plans to spend $25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be $2,712. Which of the following inequalities can be solved to find t, the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25{,}000 > (4{,}334 - 2{,}712)t$

B. $25{,}000 < (4{,}334 - 2{,}712)t$

C. $25{,}000 - 4{,}334 > 2{,}712t$

D. $25{,}000 > \dfrac{4{,}332}{2{,}712}t$